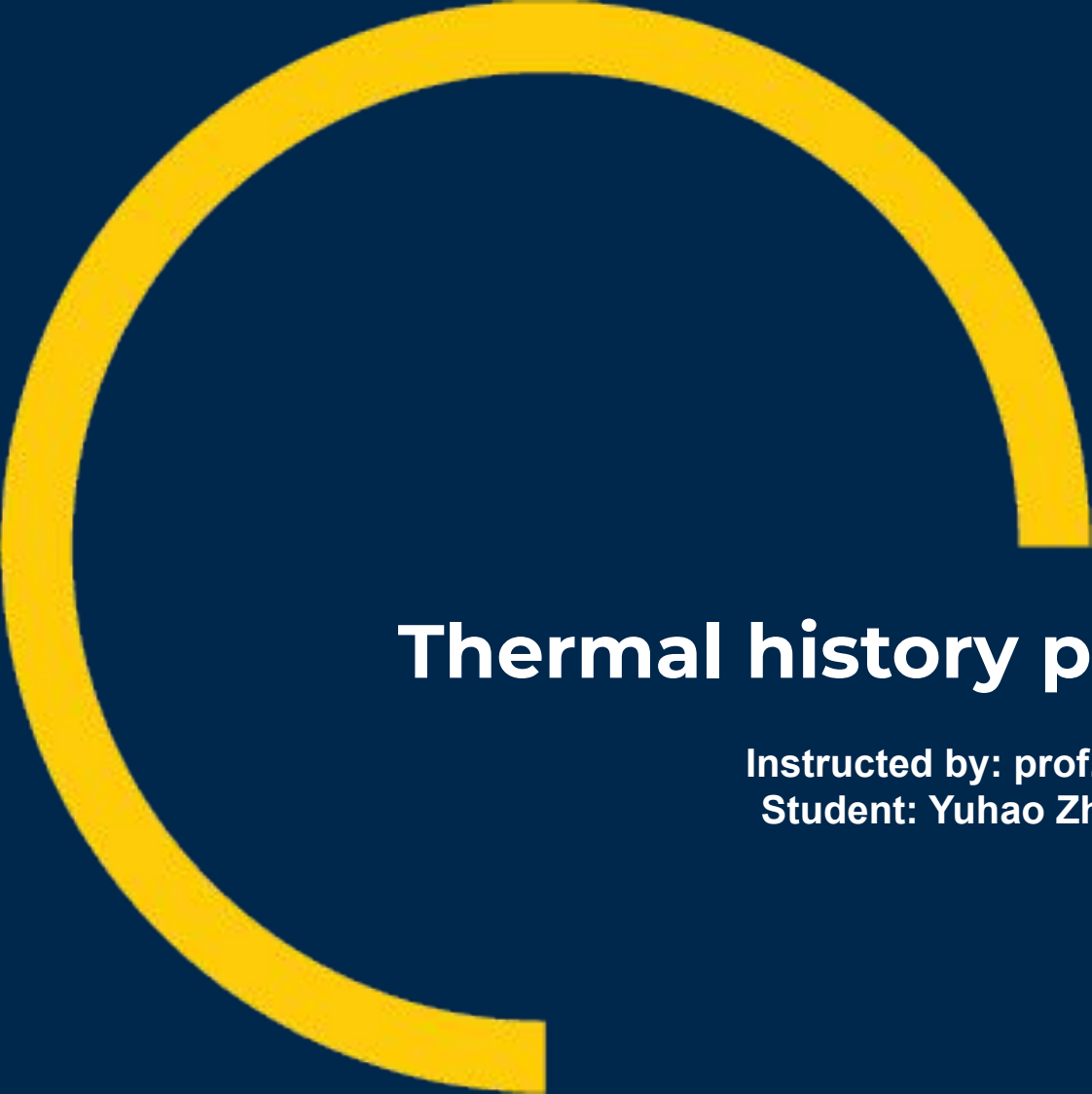




Thermal history prediction of LPBF

Instructed by: prof.Tan
Student: Yuhao Zhou

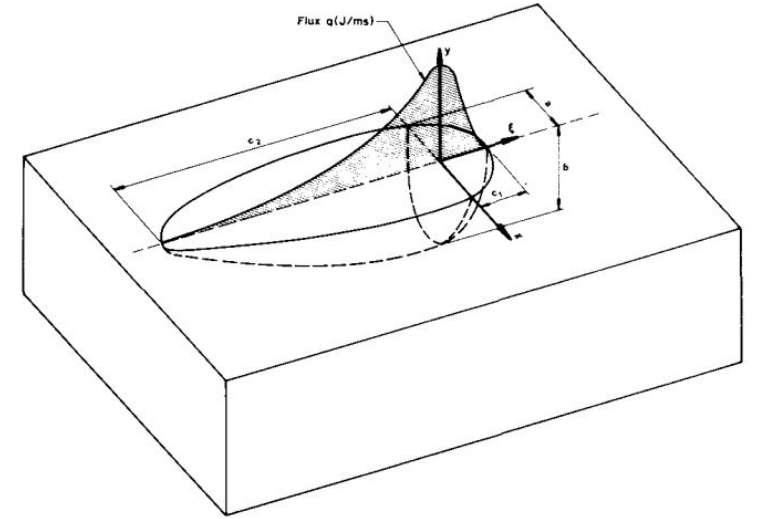
1. Review of analytical model of heat transfer of moving heat source
2. Result and runtime analysis of analytical model of complex scanning path
3. Coupling of serial version of analytical model and CA model
4. Optimization of CA code
5. Result of FVM model of heat transfer
6. Comparison between of FVM and analytical model
7. Profiering of heat transfer model and dynamic molten pool tracking

Thermal history prediction with analytical model

heat source: dual-ellipsoid

$$q(x, y, z, t) = \frac{6\sqrt{3}f_f Q}{abc\pi\sqrt{\pi}} e^{-3x^2/a^2} e^{-3y^2/b^2} e^{-3[z+v(\tau-t)]^2/c^2}$$

$$q(x, y, z, t) = \frac{6\sqrt{3}f_r Q}{abc\pi\sqrt{\pi}} e^{-3x^2/a^2} e^{-3y^2/b^2} e^{-3[x+v(\tau-t)]^2/c^2}$$



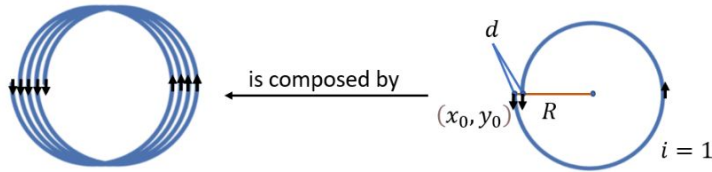
Formula:(**assume the space is semi-infinite**) detail formulation please check Me590 report

$$\theta_{Li}(x, y, z, t) = \begin{cases} 0 & (t < t_i^{\text{on}}) \\ \frac{2\eta P}{\rho C_p (\pi/3)^{3/2}} \int_0^{t-t_i^{\text{on}}} \frac{1}{\sqrt{\Phi_x \Phi_y \Phi_z}} \exp\left(-\frac{3X(t')^2}{\Phi_x} - \frac{3Y(t')^2}{\Phi_y} - \frac{3Z(t')^2}{\Phi_z}\right) dt' & (t_i^{\text{on}} \leq t \leq t_i^{\text{off}}) \\ \frac{2\eta P}{\rho C_p (\pi/3)^{3/2}} \int_0^{t_i^{\text{off}}-t} \frac{1}{\sqrt{\Phi_x \Phi_y \Phi_z}} \exp\left(-\frac{3X(t')^2}{\Phi_x} - \frac{3Y(t')^2}{\Phi_y} - \frac{3Z(t')^2}{\Phi_z}\right) dt' & (t > t_i^{\text{off}}) \end{cases}$$

where, $\Phi_i = 12\kappa(t - t_i^{\text{on}} - t') + \sigma_i^2$ for $i = x, y, z$

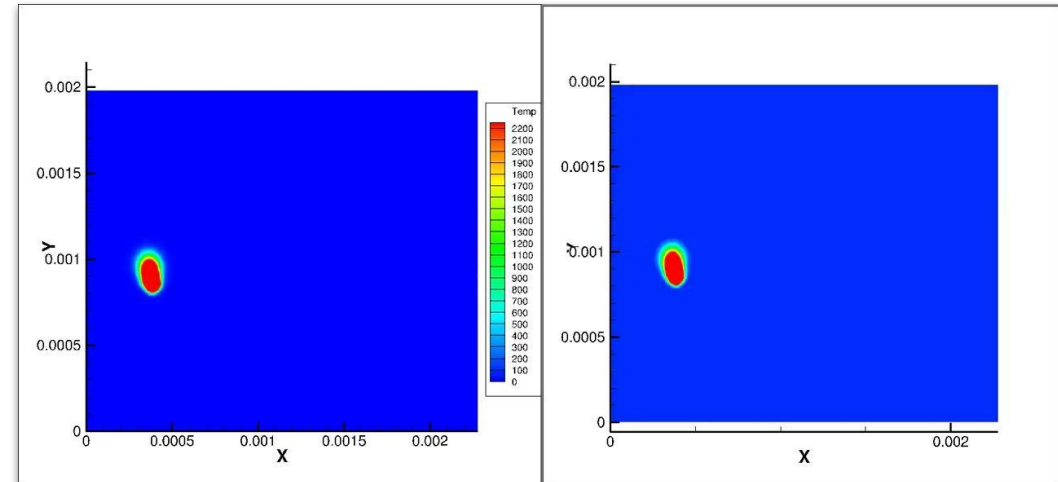
Thermal history prediction of complex scanning strategy

- Commonly-used iterative method for heat transfer such as FVM is **computational expensive**.
- Analytical and semi-analytical model can be used based on few simplifications of conduction model.
- Example: spiral scanning path**

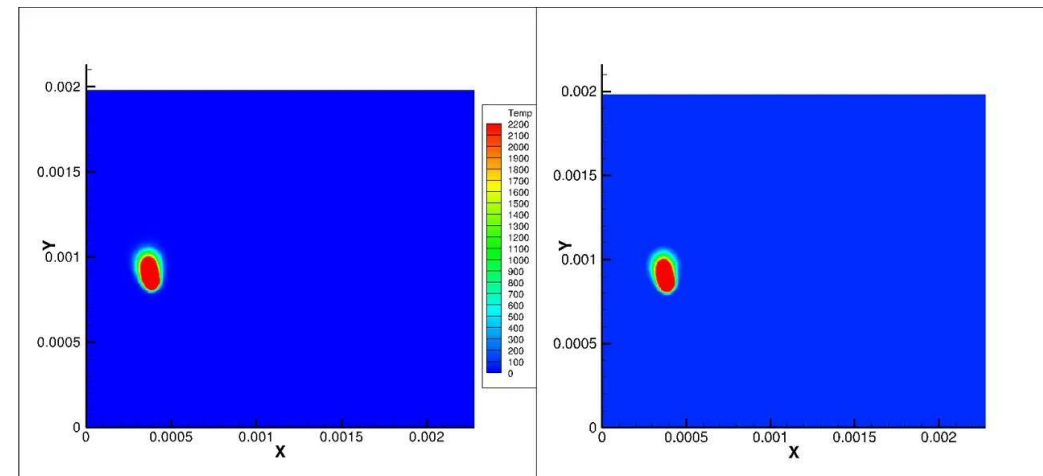


- Area of interest:
 - Length: 1.5×10^{-3} (m)
 - Width: 1.5×10^{-3} (m)
 - Height: 1×10^{-3} (m)
 - Mesh size: 2×10^{-4} (m)
- Scanning parameters:
 - can be extracted from discretized data
 - R: $0.63628482 \times 10^{-3}$ (m)
 - d: $0.10194418 \times 10^{-3}$ (m)
 - time of each spiral cycle: 4.04824×10^{-3} (s)

5 spiral cycles



10 spiral cycles



Microstructure prediction: coupling analytical thermal with CA

- **Coupling thermal model and CA model**

- Thermal history of complex scanning strategy could be computed using analytical and semi-analytical model, then imported into the microstructure prediction model.
- During solidification, dendrite growth and competition is modeled using cellular automata (CA) algorithm.

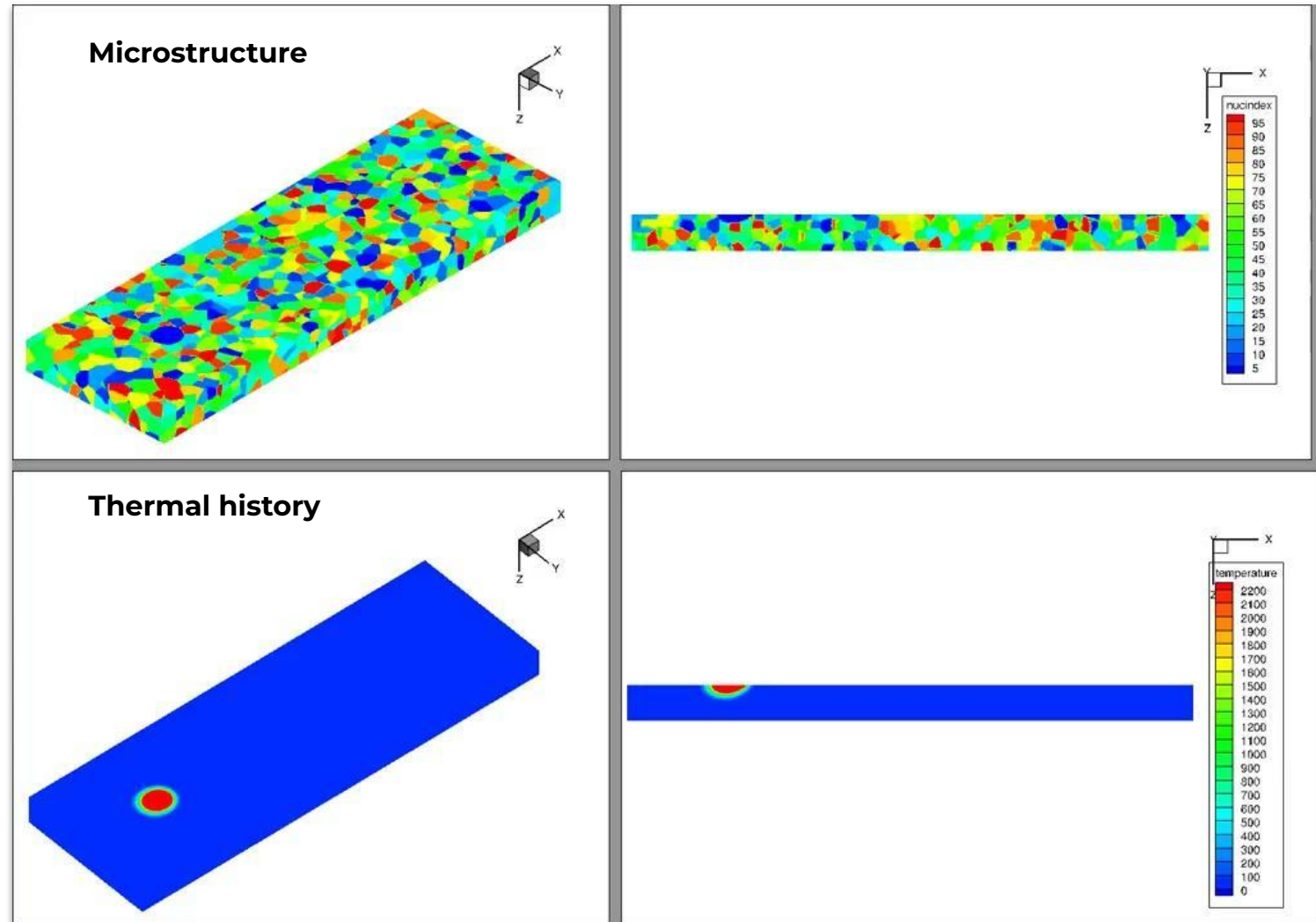
- **Example of a single track:**

- Area of interest:

- Length: 1.5×10^{-3} (m)
- Width: 5×10^{-4} (m)
- Height: 1×10^{-4} (m)
- Mesh size: 3×10^{-6} (m)

- Time of interest:

- Time step: 3×10^{-7} (s)
 - calculated by $0.1 * dx / \text{solidification speed}$
- Number of time step: 5400



Optimization of CA – discretize angle orientation

To accelerate CA model, discretize angles and their sin cos function in octahedron calculation

Previous version:

- generate continuous double float randomly
- whenever using angles, cos or sin is re-performed

for t,i,j,k:

{

- 3 angles defined in main function:

```
Phi[i][j][k] = acos(2. * rand() / (1. * RAND_MAX - 1.)) * 180. / PI;
```

```
phi2[i][j][k] = (360. * rand() / 1. * RAND_MAX);
```

```
phi1[i][j][k] = (360. * rand() / 1. * RAND_MAX);
```

- used in function

Crystal_to_Sample:

```
cosine = cos(Phi * PI /
```

```
180); sine = sin(Phi * PI /
```

```
180); cosine = cos(phi1 * PI
```

```
/ 180); sine = sin(phi1 * PI /
```

- used in function Sample_to_Crystal:

```
180); cosine = cos(Phi * PI /
```

```
180); sine = sin(Phi * PI /
```

```
180); cosine = cos(phi2 *
```

```
PI / 180); sine = sin(phi2 *
```

```
PI / 180);
```

} complexity of cos, sin:

$O(t * i * j * k)$

Updated version:

- pre-calculate discrete list of angle and corresponding cos and sin
- directly randomly generate discrete sin and cos for 3 angles

and pass to functions that uses these angles

- out of loop, pre-calculate uniform discrete list sin and

```
COS angleList = {0, ..., 2*PI}; // with step = 2 * PI /
```

```
numAngleClass
```

- sin and cos list of phi1 and

```
phi2 sinListphi = {sin(0), ... , sin(2*PI)}; // same for
```

```
cosListphi
```

- sin and cos list of

```
Phi sinListPhi = {sin(0-PI), ... , sin(PI)}; // same for
```

```
cosListPhi
```

for t,i,j,k:

{

- randomly define cos and sin of 3 angles in main

```
function PhiIndex = rand() % numAngleClass; // generate a random integer
```

```
index PhiSin[i][j][k] = sinListPhi [ PhiIndex ]; // directly call the sin
```

```
value of Phi
```

- directly pass sin and cos of 3 angles to the function and use them

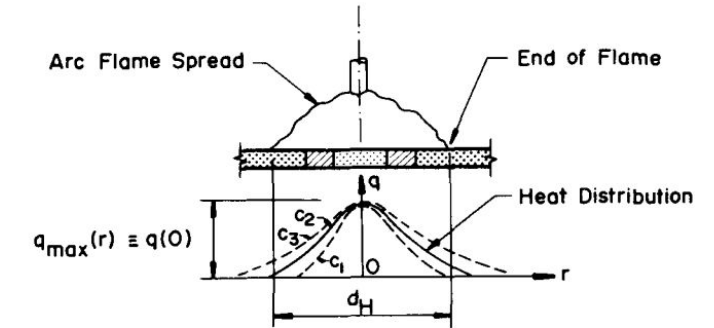
} complexity of cos, sin:

$O(\text{numAngleClass})$

Thermal history prediction with FVM model

heat source: Gaussian surface flux distribution

$$q(x, \xi) = \frac{3Q}{\pi c^2} e^{-3x^2/c^2} e^{-3\xi^2/c^2}$$



$$c_1 > c_2 > c_3$$

$d_H \equiv$ Hot Spot (Dia.)

$c \equiv$ Concentration Coefficient

Formula:

$$\Gamma_e A_e \left(\frac{\phi_E - \phi_P}{\delta x_{PE}} \right) - \Gamma_w A_w \left(\frac{\phi_P - \phi_W}{\delta x_{PE}} \right) + [\Gamma_n A_n \left(\frac{\phi_E - \phi_P}{\delta x_{PE}} \right) - \Gamma_s A_s \left(\frac{\phi_P - \phi_W}{\delta x_{PE}} \right)] + [\Gamma_t A_t \left(\frac{\phi_E - \phi_P}{\delta x_{PE}} \right) - \Gamma_b A_b \left(\frac{\phi_P - \phi_W}{\delta x_{PE}} \right)] + (S_u + S_p \phi_p) = 0$$

Thermal model using FVM

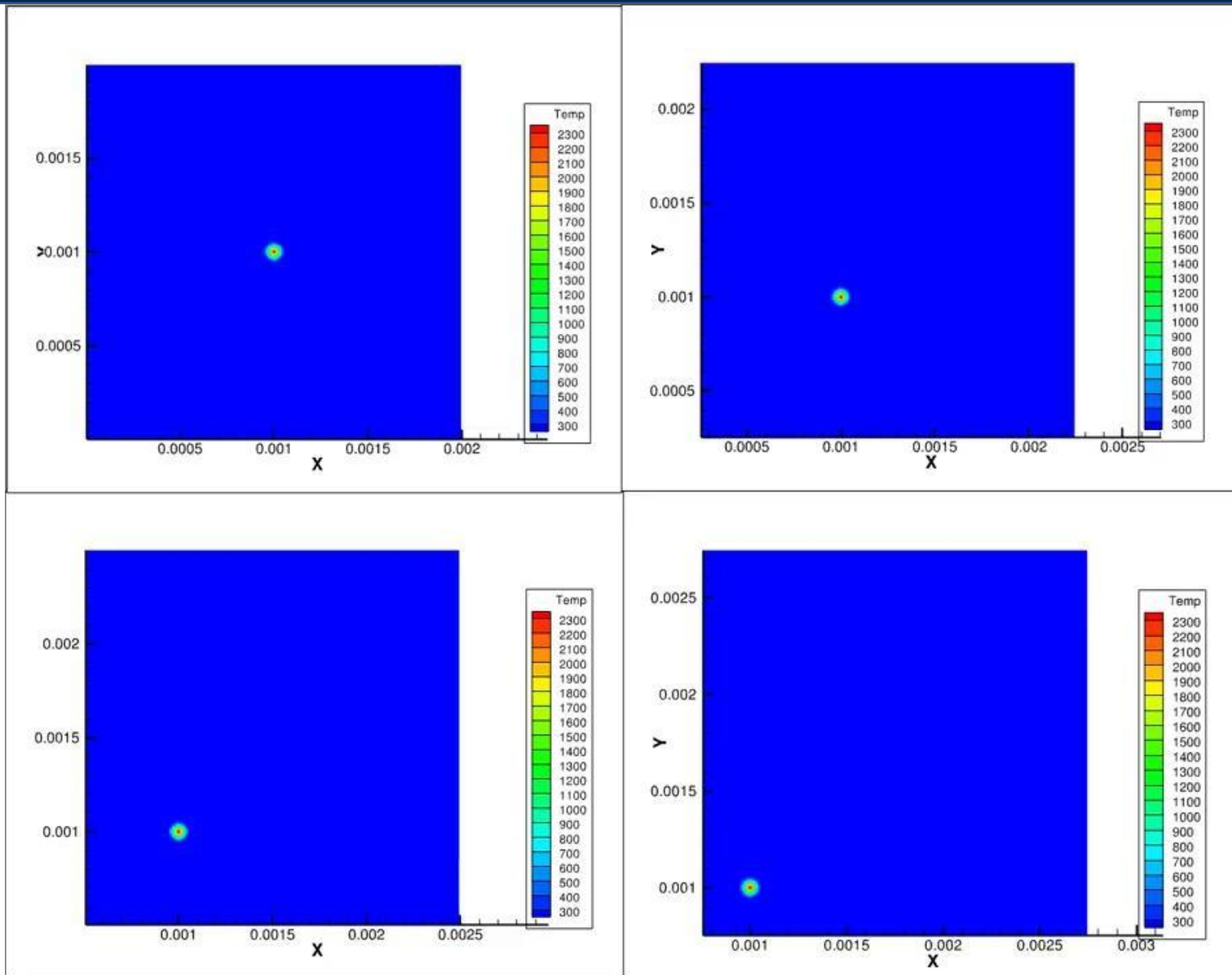
Stationary spot modeled using FVM:
molten pool “distortion”

```
Material parameters list:
Material name:      steel
Density:           7780
Power absorptivity: 0.3
Specific heat:      523
Thermal conductivity: 20
Thermal diffusivity: 4.91529e-06
Liquid temperature: 1941

Cube parameters list:
Len (x/y/z):       0.002 / 0.002 / 0.001
Start (x/y/z):     0 / 0 / 0
Mesh size:         1e-05
Mesh num (x/y/z):  200 / 200 / 100

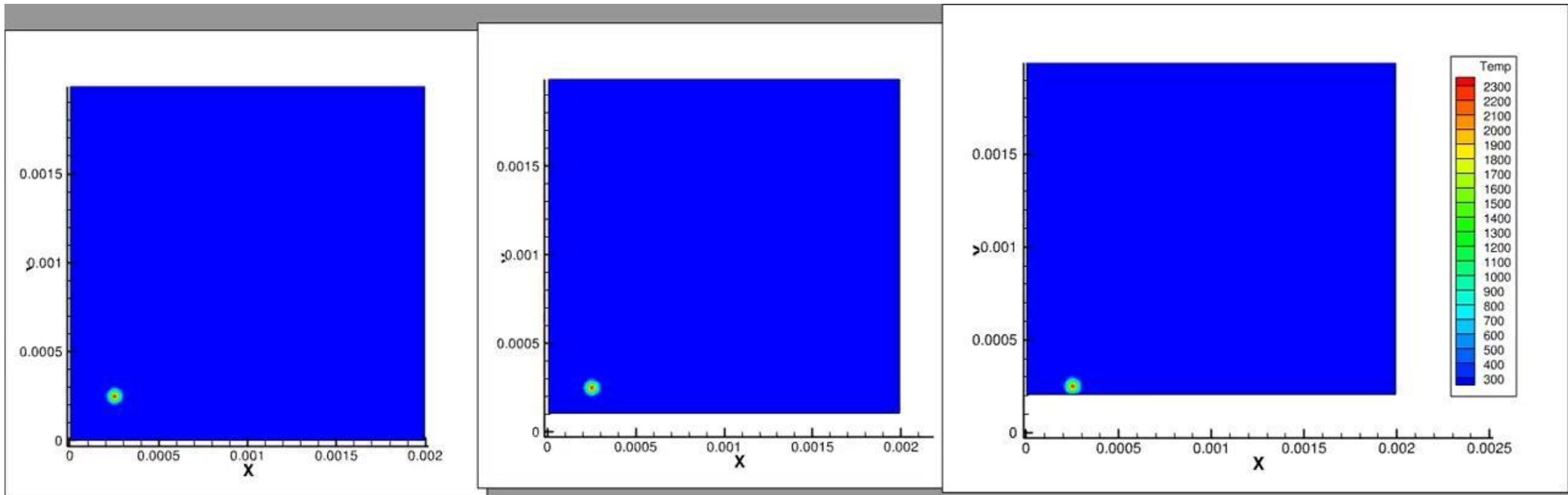
Laser parameters list:
Laser power:       280
Beam radius of X:  5e-05
Beam radius of Y:  5e-05
Beam radius of Z:  5e-06
Scan time:         0.005
Scan pattern:      0

Time parameters list:
Total time:        0.007
Time step:         2.71263e-06
Num time step:     2581
```



Thermal model using FVM

Single track scan modeled using FVM: molten pool “distortion”

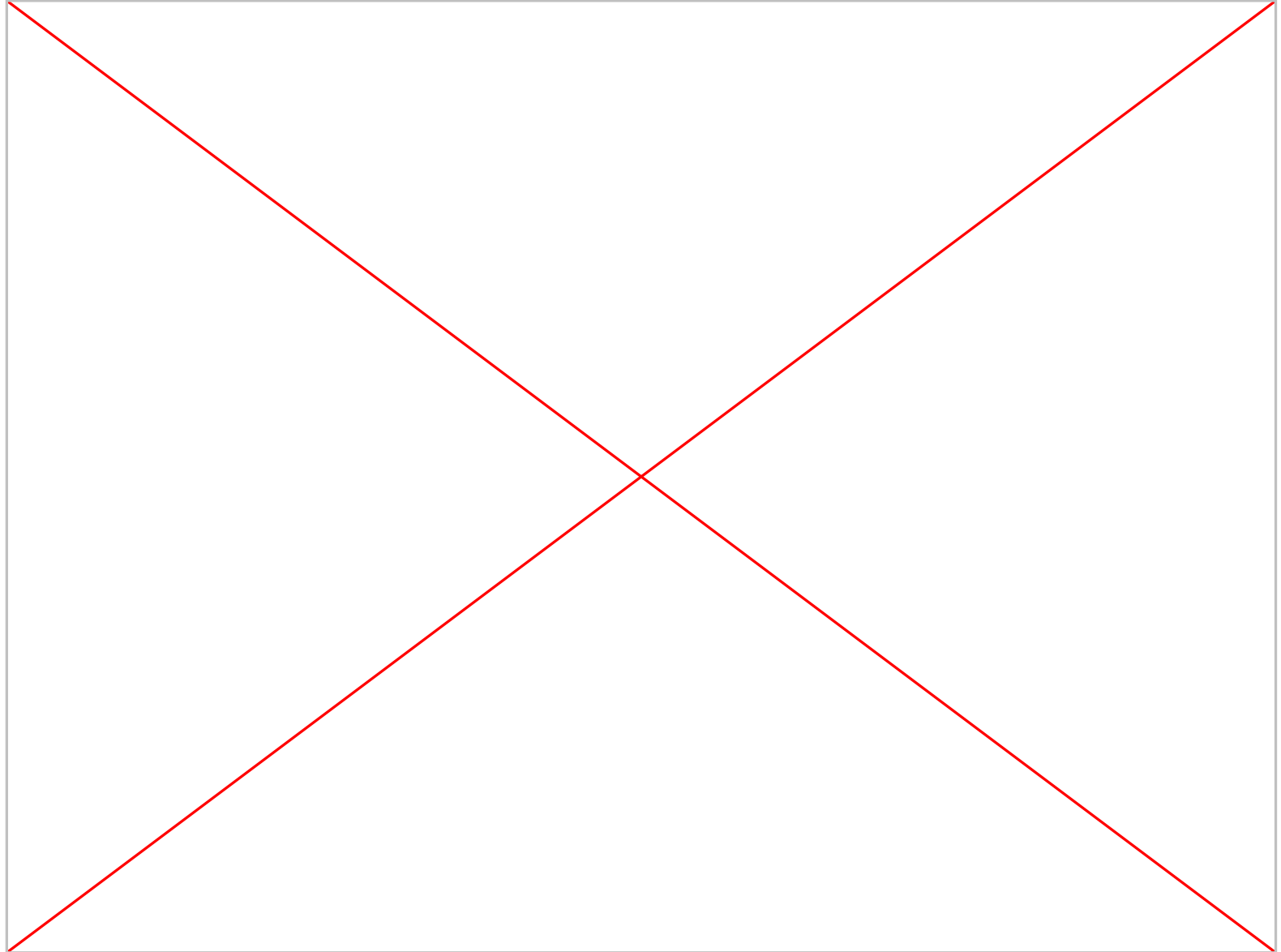


Comparison: analytical model and FVM model

3 figures above are FVM model results.

3 figures below are analytical model results.

- **Analytical model** is based on **semi-infinite space** for heat conduction, therefore, the temperature field of single track at different place is “translational”. **The molten pool will not be affected by the boundary.**
- **FVM result is more sensitive to the position of scan track with respect to the boundary.**



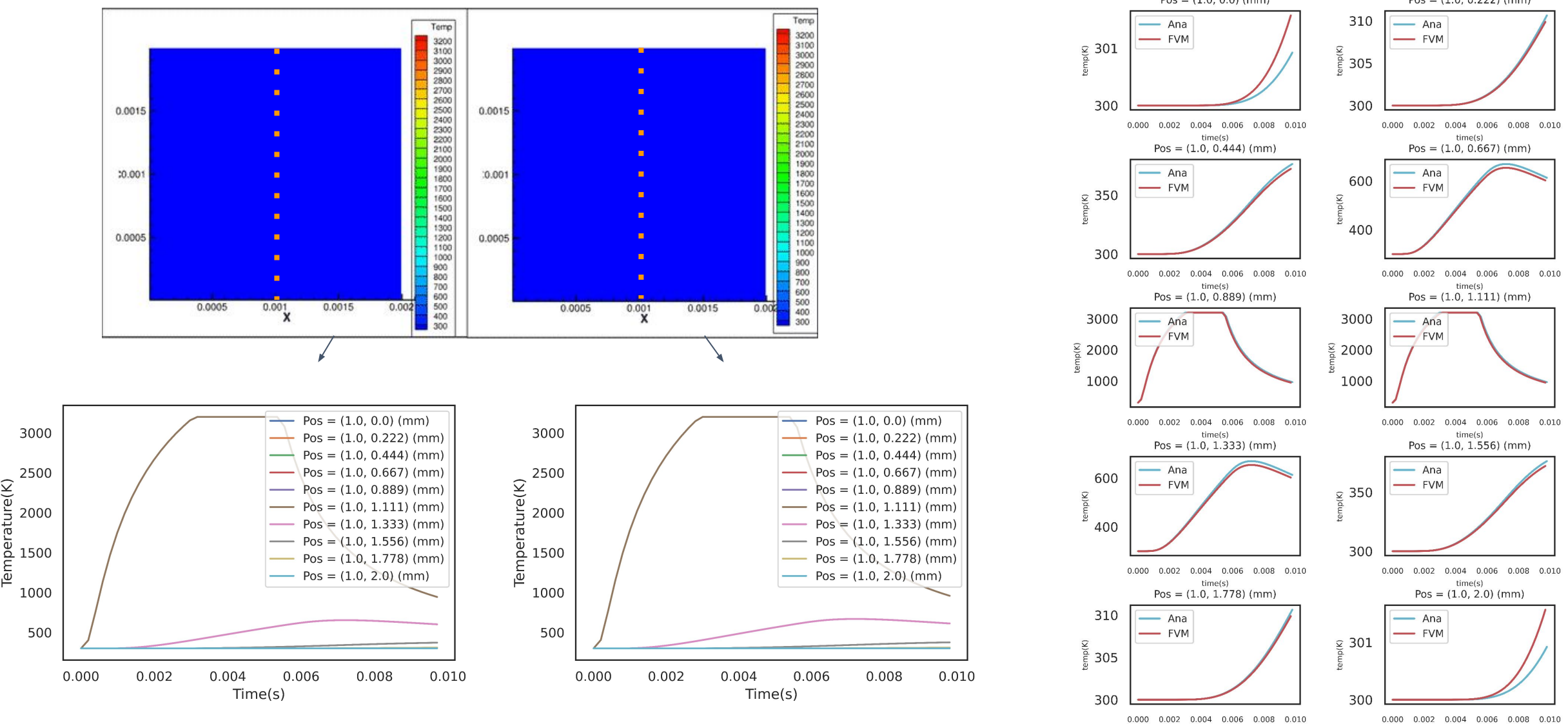
Comparison: analytical and FVM model - stationary at center



Analytical method

FVM method

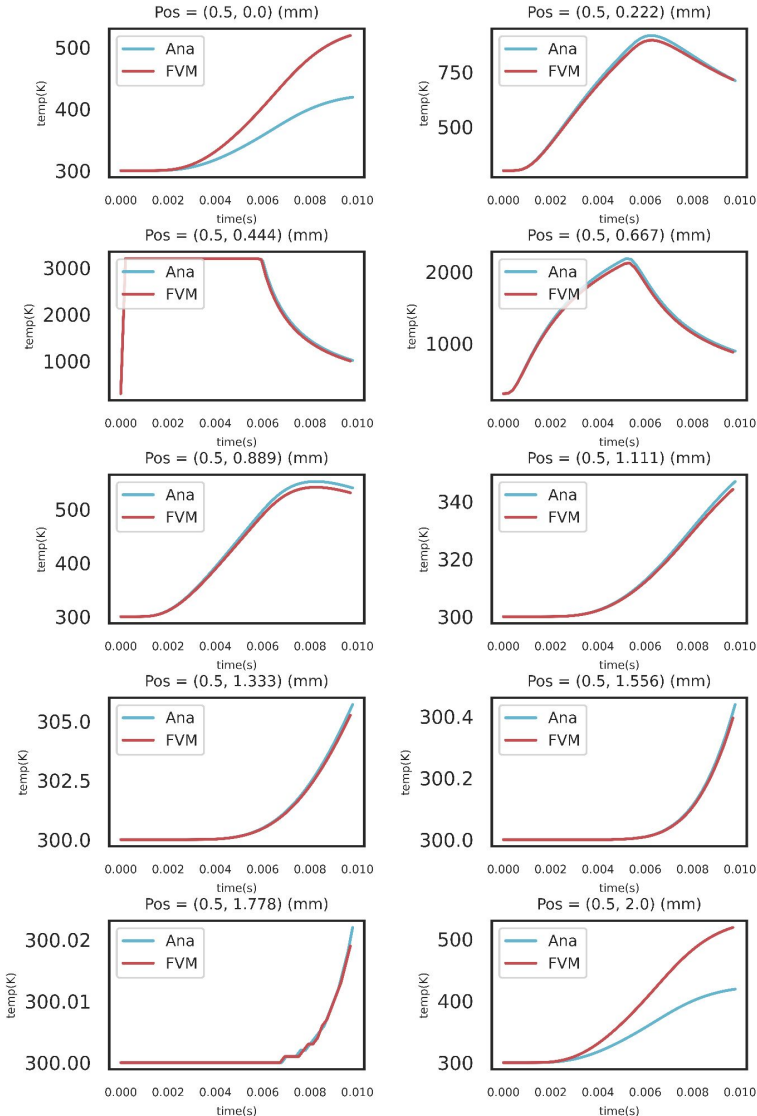
Compare two method of single point



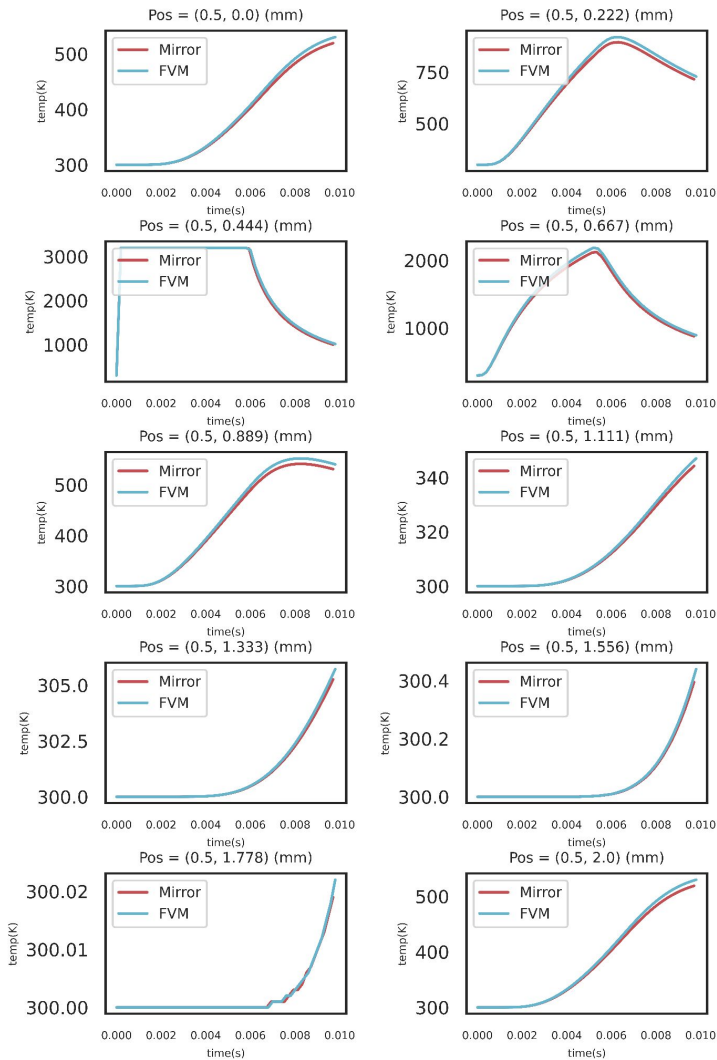
Comparison: analytical and FVM model - stationary near corner



Ana vs FVM



FVM vs Mirror

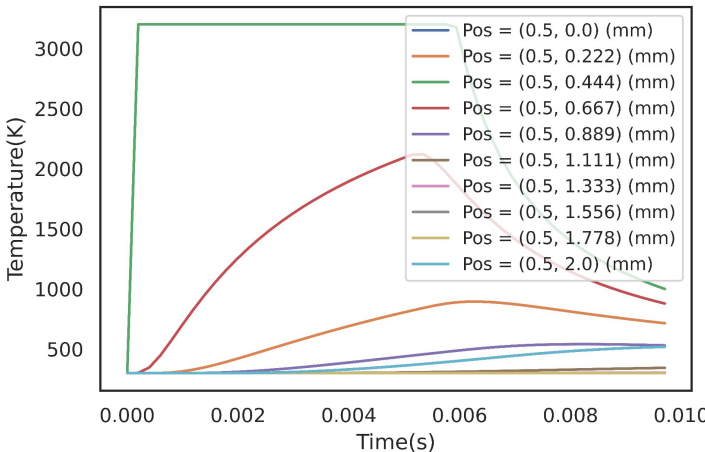
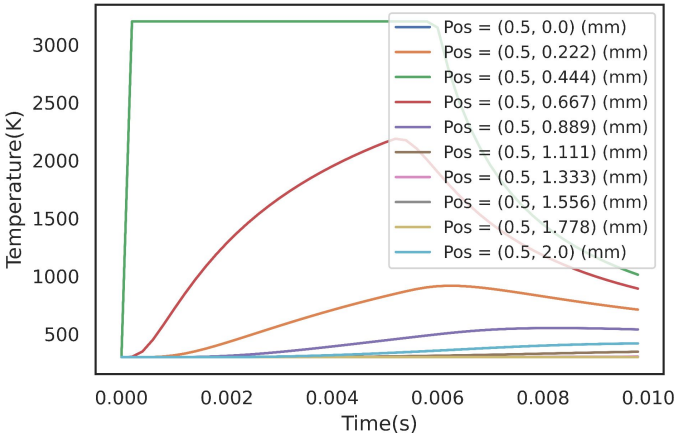
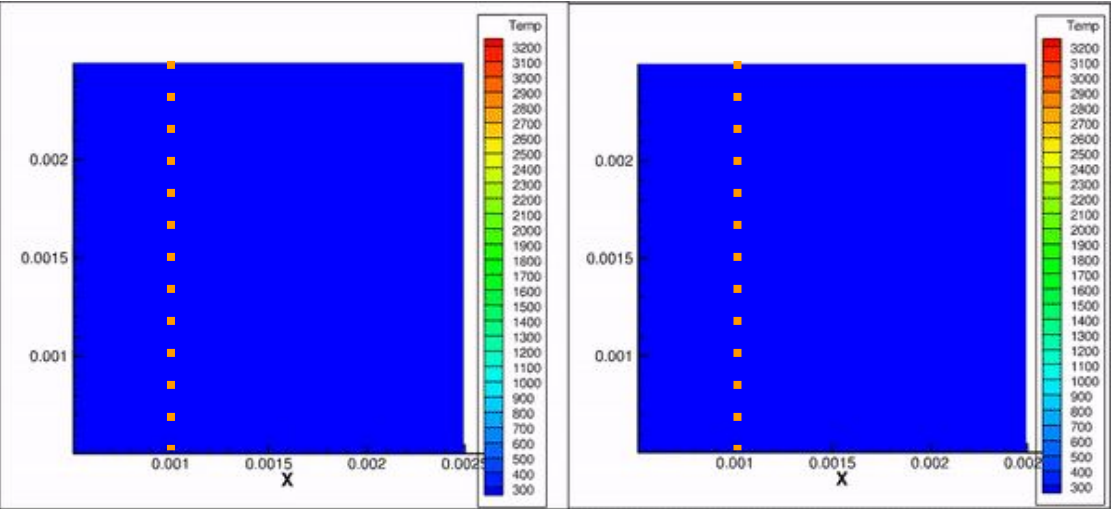


Comparison: analytical and FVM model

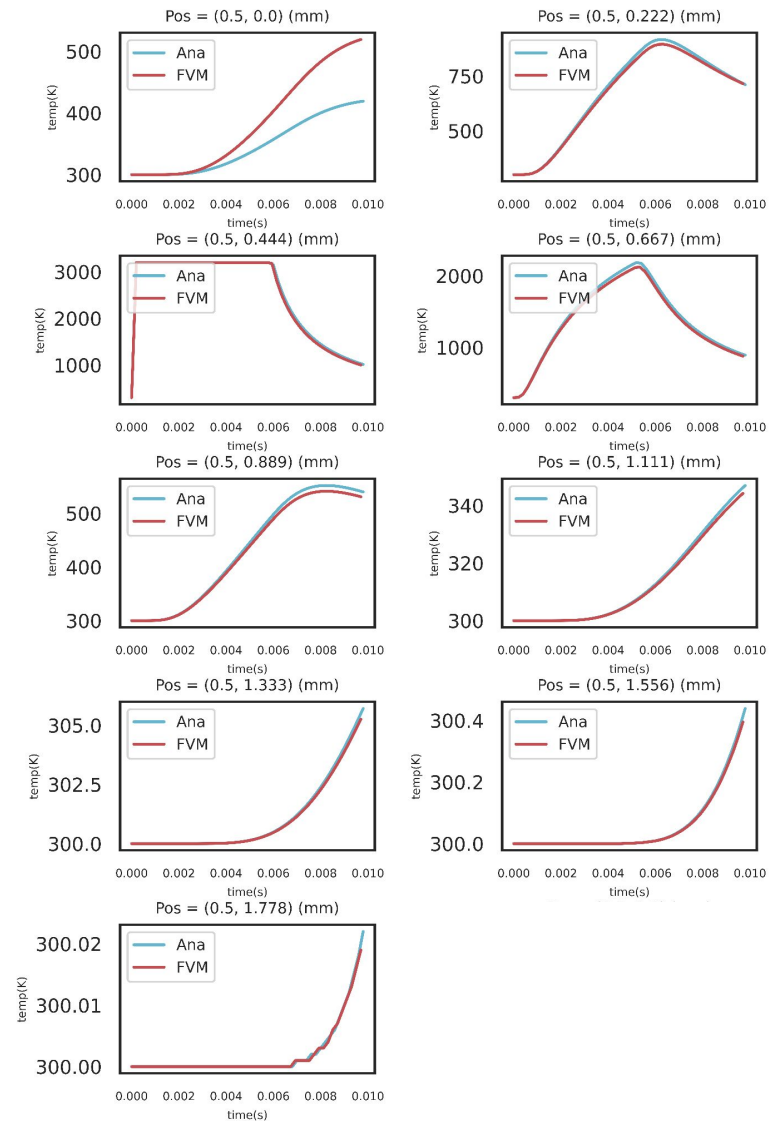


Analytical method

FVM method



Compare two method of single point



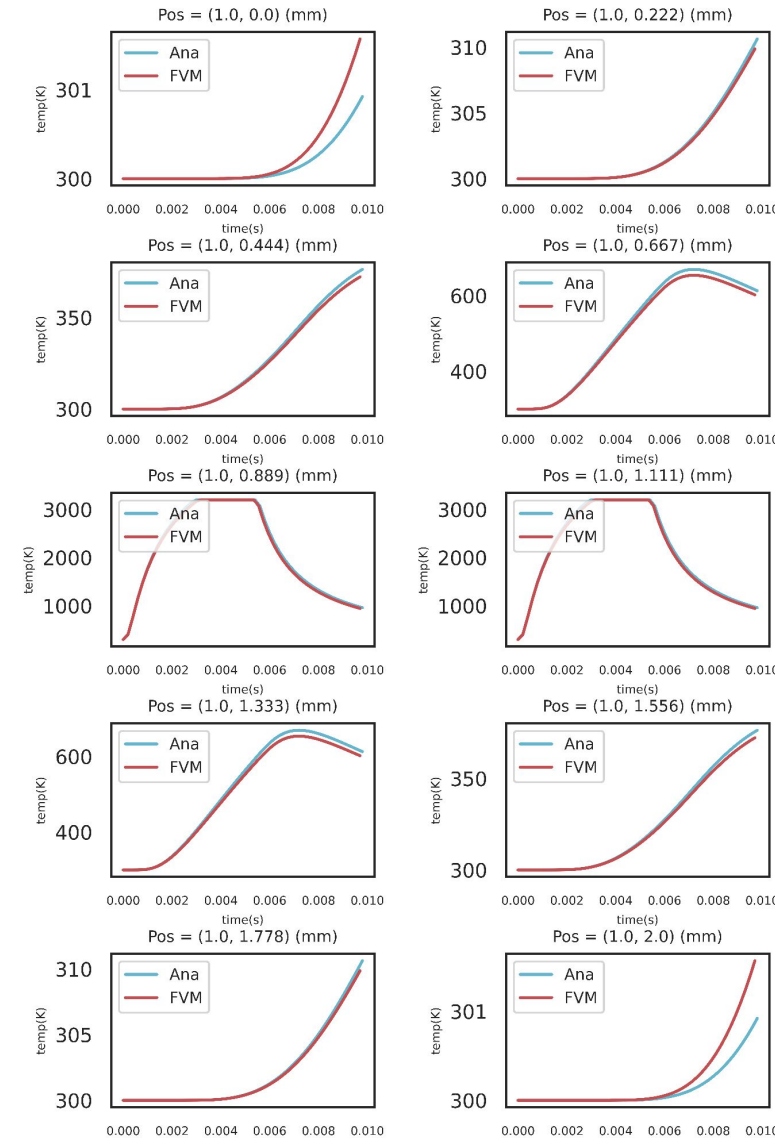
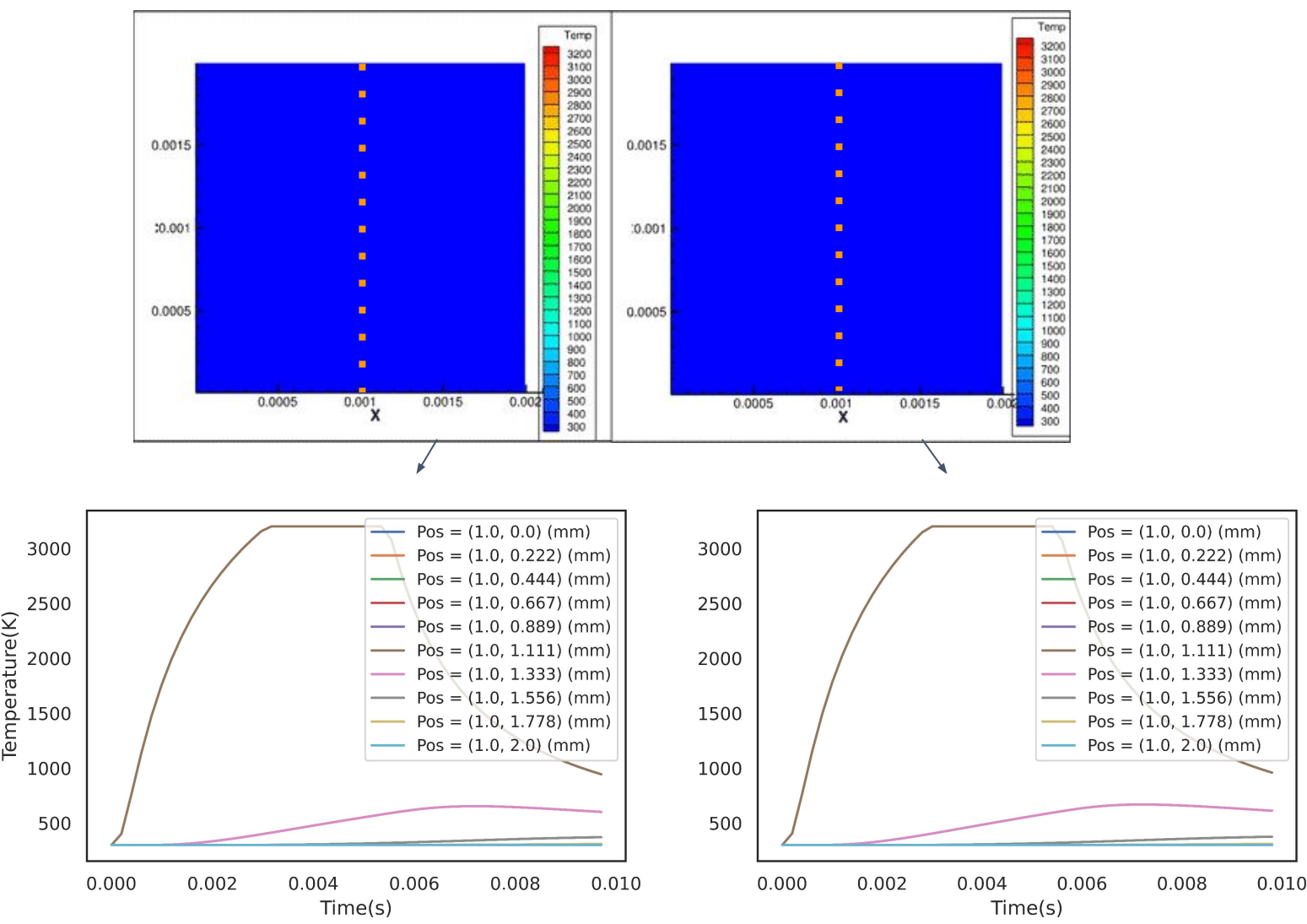
Comparison: analytical and FVM model - linear at center



Analytical method

FVM method

Compare two method of single point



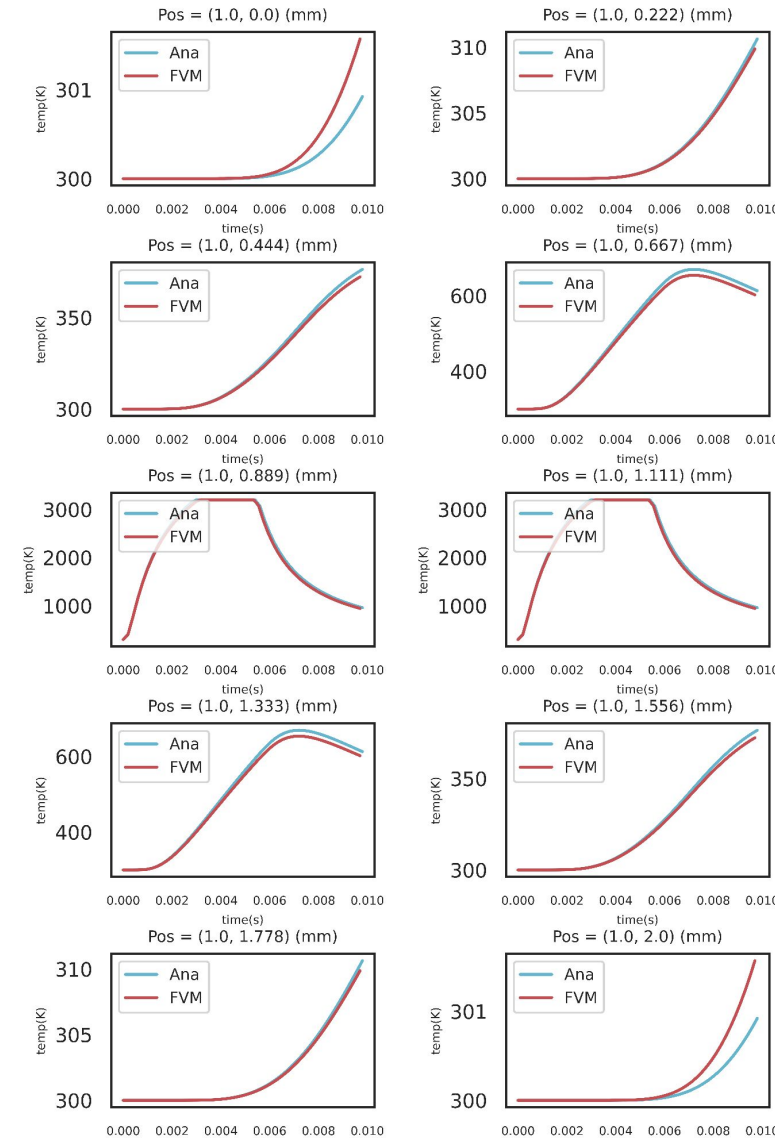
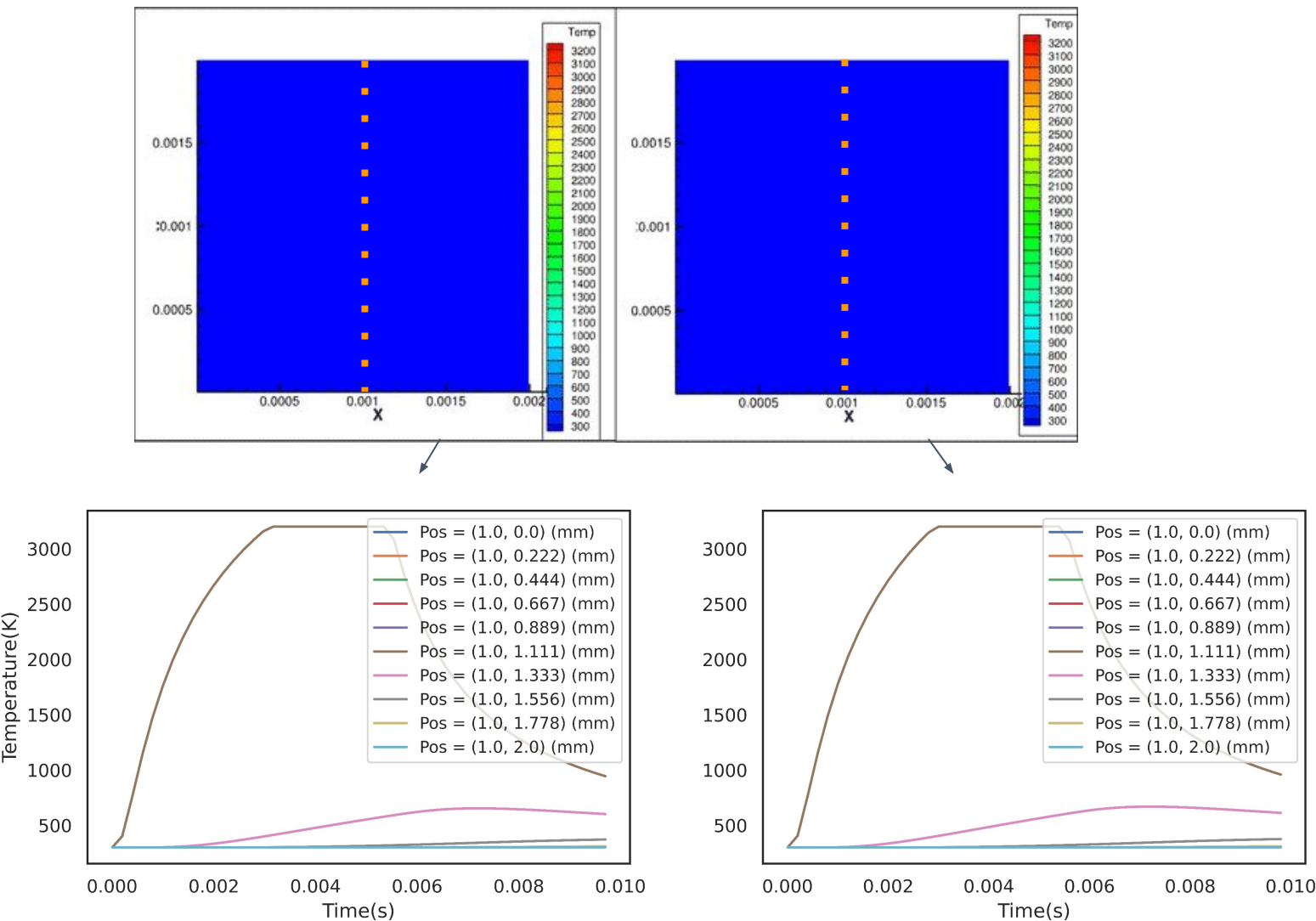
Comparison: analytical and FVM model - linear 0.5mm to edge



Analytical method

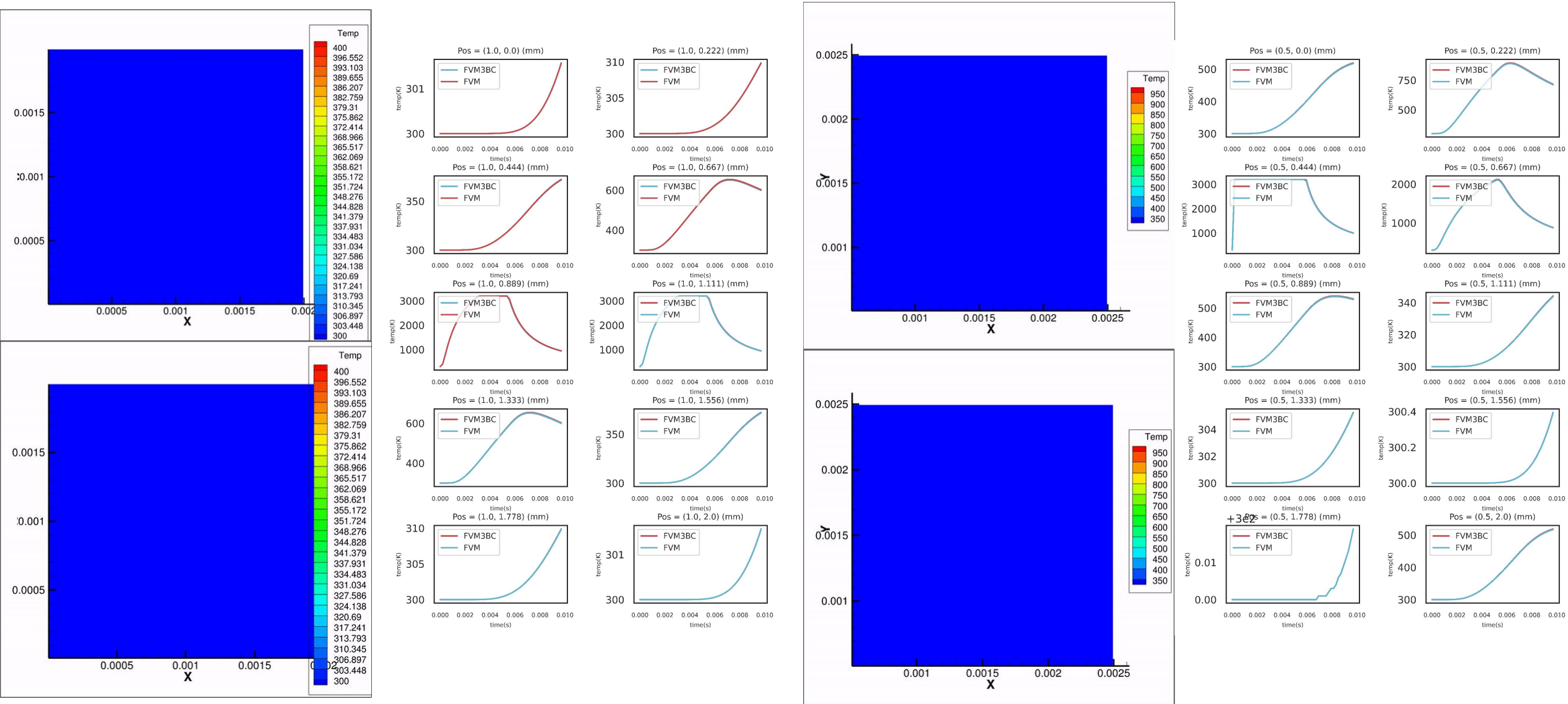
FVM method

Compare two method of single point



FVM model with convective and radiative BC - nearly no effect

- simulation result



FVM model with convective and radiative BC - nearly no effect – explanation

$$-k\nabla T = h(T - T_0) + \sigma\epsilon(T^4 - T_0^4)$$

$$\Rightarrow -k \left(\frac{T_p - T_b}{\delta_x} \right) = h(T - T_0) + \sigma\epsilon(T^4 - T_0^4)$$

Solve for T_b , transform mixed BC to 1st BC
for uniform grid case:

$$\Rightarrow \text{Find the root for } f(T_b) = 0$$

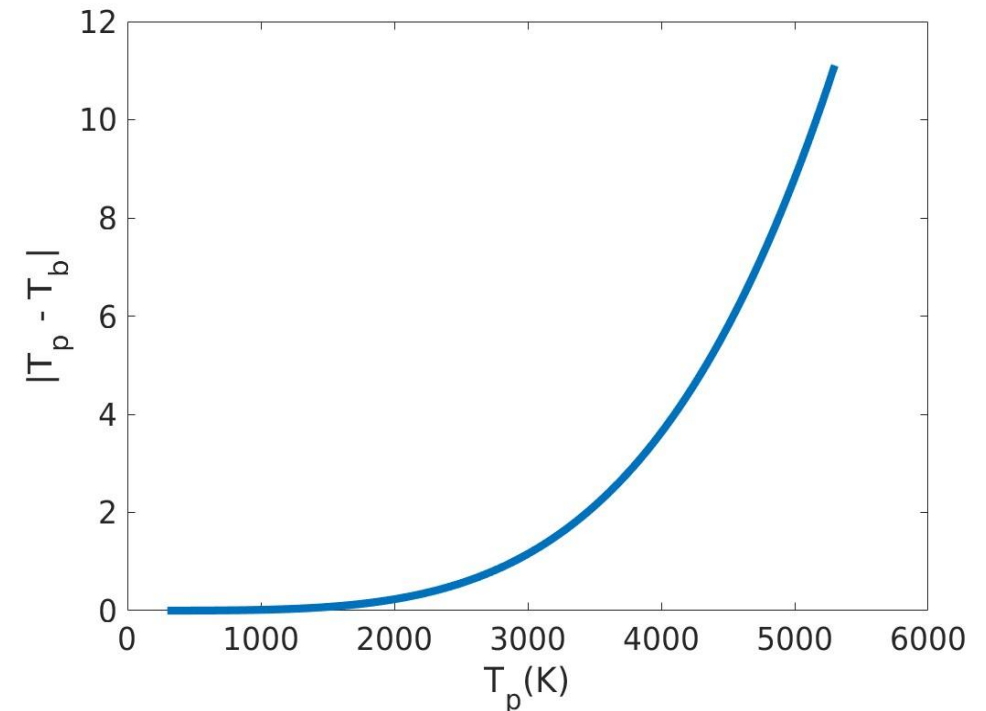
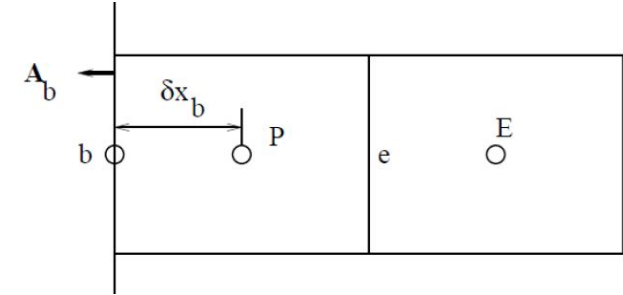
$$= \sigma\epsilon T_b^4 + \left(h + \frac{2k}{\Delta x} \right) T_b + \left(\sigma\epsilon T_0^4 - hT_0 - \frac{2k}{\Delta x} T_p \right)$$

where, $T_b \in [T_0, T_p]$

With T_p changes dramatically between ambient temp and metal vapor temp

$|T_b - T_p|$ changes very little compared with T_p ,

Therefore, the thermal diffusion caused by the convection and radiation will be very small.



Verification conclusion:



For a short time heat transfer simulation, analytical model almost produce the same result of iterative method as FVM when it is far away from the boundary.

When the laser scanning track is near the boundary, simple “Mirroring” the thermal field outside the valid build volume calculated based on semi-infinite assumption back into the build volume would also effectively increase the accuracy of the temperature field. (method of imaging is accurate enough for fast predicting process)

Runtime analysis: profile analytical thermal model

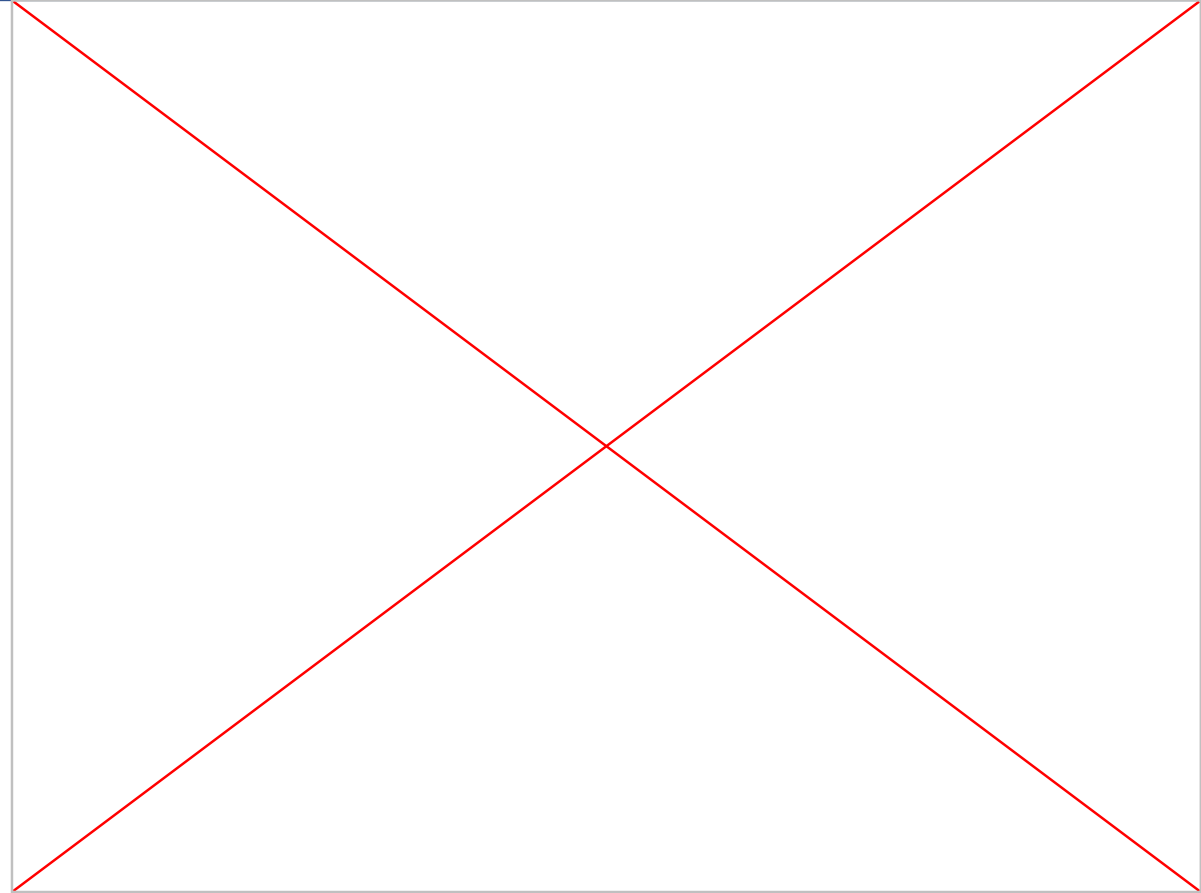
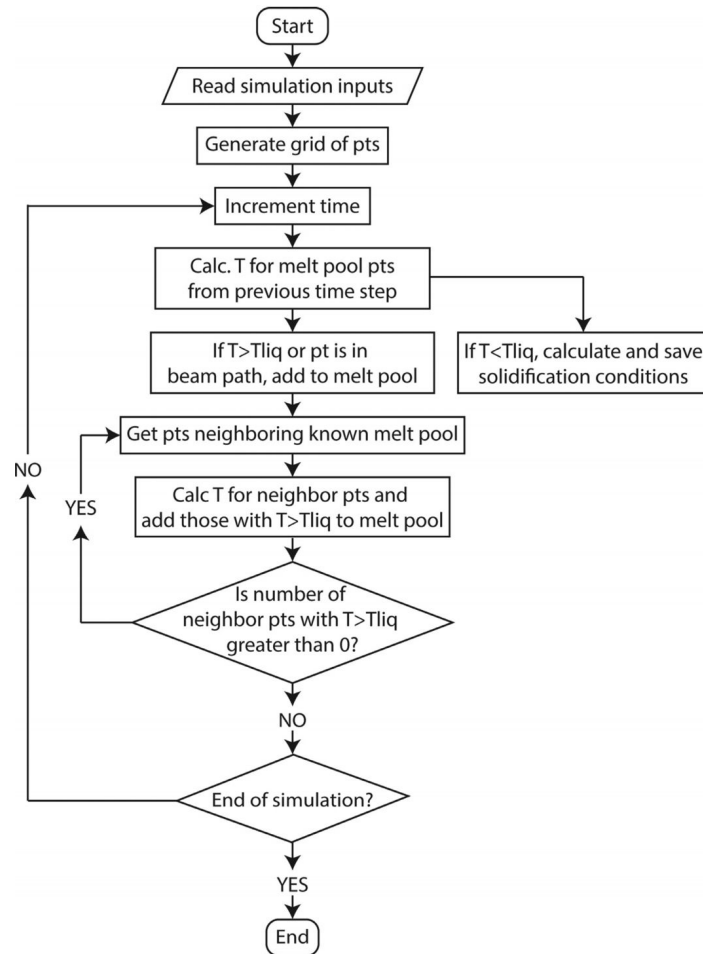
Runtime bottleneck of analytical solution: numerical integration

At each time step, for each point of interest, numerical integration is quite expensive

Method	Samples
✓ 93.4% heatSpiral`_start	31,294
✓ 93.4% libc.so.6`_libc_start_main@@GLIBC_2.34	31,294
✓ 93.4% libc.so.6`_libc_start_call_main	31,294
✓ 93.4% heatSpiral`main	31,294
✓ 92.2% heatSpiral`computeThermalFieldWithGaussianLegendre	30,906
✓ 92.1% heatSpiral`summationOfGaussianLegendre	30,869
✓ 82.3% heatSpiral`calculateIntegrandVal	27,581
> 43.1% libm.so.6`pow@@GLIBC_2.29	14,451
> 14.0% libm.so.6`exp@@GLIBC_2.29	4,695
2.1% heatSpiral`[unknown]	718
< 1% libm.so.6`_sqrt	302
< 1% libm.so.6`_sqrt_finite@GLIBC_2.15	13
> < 1% [kernel.kallsyms]`asm_sysvec_apic_timer_interrupt	5
< 1% nvidia.ko`_nv033075rm	1
< 1% [kernel.kallsyms]`error_entry	1
> < 1% [kernel.kallsyms]`asm_sysvec_apic_timer_interrupt	4
< 1% [kernel.kallsyms]`sync_regs	1
< 1% heatSpiral`calculateIntegrandVal	10
< 1% libgsl.so.27.0.0`gsl_integration_glfixed_table_alloc	7
> < 1% libc.so.6`printf	4
> 1.2% heatSpiral`writeDatFirstLayer	386
< 1% heatSpiral`summationOfGaussianLegendre	2
> 5.9% [unknown]`[unknown]	1,978
> < 1% [stack]`[unknown]	227
> < 1% heatSpiral`[unknown]	7
> < 1% ld-linux-x86-64.so.2`_GLOBAL_OFFSET_TABLE_	3
< 1% [kernel.kallsyms]`asm_sysvec_apic_timer_interrupt	2
> < 1% [kernel.kallsyms]`entry_SYSCALL_64_after_hwframe	1
> < 1% [heap]`[unknown]	1

Dynamic molten pool tracking algorithm implementation

Molten Pool Tracking Algorithm



This algorithm can be implemented to track the thermal field from above melt temperature downwards to any customized temperature contour, which can dramatically reduce computation expense when ratio (molten pool volume / total build volume) is small. For this single case, if we only track temperature dynamically within range [3200, 4500] K, it would make runtime 30 times faster than traditionally analytical method.